

SANDESH POUDEL
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PROFESSIONAL SUMMARY – DATA ENGINEER

- 7+ years of experience as a **Data Engineer**, designing and optimizing large-scale **ETL/ELT pipelines, real-time data processing systems, and cloud-based** data platforms across Azure, AWS, and GCP.
- Expert in **Apache Spark (Scala, PySpark), Kafka, Airflow, and Databricks** for processing petabyte-scale structured and semi-structured data in both batch and streaming environments.
- **Built and orchestrated pipelines using Azure Data Factory, AWS Glue, and GCP Dataflow**, enabling seamless data movement and transformation across multi-cloud systems.
- Migrated legacy data systems to **Snowflake, Big Query, and Azure Synapse Analytics**, improving query performance and reducing reporting latency by up to 30%.
- Implemented **Medallion architecture** (Bronze, Silver, Gold layers) in **Azure Data Lake**, ensuring data reliability and scalability for advanced analytics and machine learning.
- Led containerization of **ETL pipelines using Docker and Kubernetes**, and automated infrastructure provisioning with Terraform to ensure robust and scalable deployment.
- **Developed interactive dashboards with Power BI and Tableau**, using complex DAX functions and custom Power Apps to deliver actionable business insights.
- Designed and implemented **real-time data ingestion workflows using Apache Kafka, Spark Streaming, and FHIR APIs** for immediate healthcare analytics.
- Performed **SQL query optimization**, data modeling (Star and Snowflake schema), and created Fact and Dimension tables using tools like Erwin to support BI and reporting.
- **Integrated diverse data sources using Informatica PowerCenter, Talend, and custom APIs**, while ensuring high data quality and consistency across platforms.
- Automated CI/CD pipelines using **Jenkins, GitLab, and GitHub Actions**, supporting Agile teams and continuously deploying data solutions.
- Worked extensively with file **formats including Parquet, AVRO, XML, and Sequence Files**, supporting scalable and flexible data storage.
- **Utilized Linux/Unix, Bash scripting, and Python (Pandas, NumPy)** for automation, analytics, and data cleansing in hybrid and cloud-native ecosystems.
- Proficient in tools and platforms including **SQL Server, Oracle, Redshift, MongoDB, Hadoop (HDFS, Hive, HBase), Apache Ambari, and Prometheus**.

TECHNICAL SKILLS

- **Databases:** Snowflake, MS SQL Server, MongoDB, HBase, Hive, OLAP, AWS RDS, Oracle, MySQL
- **NoSQL Database:** MongoDB, HBase, Hive, Cassandra
- **Programming Language:** Python, Scala, SQL, Java, R, MATLAB
- **Cloud Technologies:** Azure, AWS, GCP
- **Data Formats:** CSV, JSON, Parquet, Excel, XML
- **Querying Languages:** SQL, NO SQL, PostgreSQL, MySQL, Microsoft SQL
- **CI/CD Integration Tools:** Jenkins, Git, GitLab, GitHub Actions, Azure DevOps, AWS Code Pipeline
- **Scalable Data Tools:** Databricks, Hadoop, Hive, Apache Spark, Flume, Pig, MapReduce, Sqoop, Medallion
- **Operating Systems:** Red Hat Linux, Unix, Windows, macOS,
- **Reporting & Visualization:** Power BI, Tableau, Looker, Matplotlib

PROFESSIONAL EXPERIENCE

Highmark Health
Data Engineer

Pittsburgh, PA
May 2022 – Present

Responsibilities:

- **Developed ETL pipelines** using **Spark with Scala** for data processing and **Azure Data Factory** for orchestrating these pipelines, improving data availability and reducing latency by 30%.
- **Automated and orchestrated** complex data workflows using **Apache Airflow DAGs**, achieving a 40% enhancement in operational efficiency, with **Jenkins CI/CD pipelines** facilitating the continuous integration and deployment processes
- **Migrated and optimized** enterprise data models from MS SQL Server to Snowflake, using Azure Databricks to refine ETL/ELT workflows, and Azure Synapse Analytics to extend capabilities for advanced analytics.
- Led a separate initiative to migrate healthcare data to **Google Cloud, utilizing Google Big Query and AI Platform** for enhanced predictive analytics to improve patient care outcomes.
- **Designed and implemented** ETL processes to ingest data from source systems into Azure data storage services using **Azure Data Factory, T-SQL, Spark SQL, and U-SQL**.
- Facilitated data ingestion into **Azure Data Lake, Azure Storage, Azure SQL, and Azure Synapse Analytics** with subsequent data processing in Azure Databricks.
- **Modernized legacy Hadoop workflows** by initially using **Flume and Sqoop** for batch ingestion into HDFS, then transitioning to Spark and Azure Data Factory for real-time streaming pipelines, reducing latency by 30% and enabling advanced customer behavior analytics.
- **Designed** data solutions that cut operational costs by 20% through optimized cloud resource utilization and automation, enhancing efficiency and scalability across platforms.
- Implemented Spark using Scala, leveraging Data Frames, **Spark SQL API, and Pair RDDs** for efficient data processing resulting in faster and scalable computations.
- **Employed Apache Airflow** to create, schedule, and monitor data pipelines, **building DAGs** to automate ETL processes.
- **Designed and implemented** data integration workflows and ETL pipelines using Informatica PowerCenter, enabling seamless data extraction, transformation, and loading across diverse systems and databases.
- Transitioned from Informatica Cloud for batch Salesforce CRM ETL to **Apache Kafka and Spark streaming pipelines, enabling real-time data synchronization** and reducing reporting latency by 25%.
- Implemented real-time data ingestion processes using **Apache Kafka, Spark and Scala**, enabling ingestion of petabyte-scale structured and semi-structured data reducing data latency by 20%.
- **Created** dashboards in Power BI with complex **DAX functions**, delivering actionable insights to cross-functional teams, and improving decision-making efficiency by 25%.
- Built custom Power Apps integrated with **Power BI** for interactive business applications and automation, improving operational efficiency by 15%.
- Led root cause analysis for data quality issues, collaborating with data owners to improve source accuracy and reliability.
- **Implemented** a reporting environment from the ground up, integrating multiple data sources, including **SQL and Salesforce**.
- **Deployed** and managed containerized data processing pipelines using **Kubernetes, ensuring scalability** and reliability for large-scale workloads.
- **Developed robust CI/CD pipelines using Jenkins and GitLab**, integrating seamless workflows and ensuring efficient delivery of data solutions.
- **Collaborated with Agile (Scrum) teams** to deliver solutions aligned with sprint goals, **leveraging GitHub Actions for work management and Continuous Integration/Continuous Deployment (CI/CD)**.

- **Developed real-time data pipelines** using **Apache Kafka and Spark**, improving healthcare data interoperability with FHIR APIs for immediate patient management insights.

Environment: Spark, Scala, MySQL, MS SQL Server, Azure Data Factory, SQL, Apache Airflow, DAG, Azure Databricks, Azure Synapse Analytics, Flume, Sqoop, Pig, MapReduce, Informatica PowerCenter, Cloud Computing platform, FHIR APIs, Google AI Platform, Snowflake, Power BI, DAX, Salesforce, Apache Kafka, Power Apps, Kubernetes, GitLab, GitHub Actions, Jenkins.

BOK Financial
Data Engineer

Tulsa, OK
Jun 2019-Mar 2022

Responsibilities:

- **Developed** high-performance Spark code using **PySpark, Scala, and Spark-SQL/Streaming**, optimizing large-scale data transformations and workflows within the Hadoop ecosystem.
- **Structured** data into **Medallion architecture layers (Bronze, Silver, Gold)** using **Azure Data Lake**, ensuring high data quality and scalability for advanced analytics.
- **Designed** multi-cloud ETL pipelines using **AWS Glue for native AWS workloads and Matillion** for hybrid integrations with Azure and GCP, ensuring seamless data interoperability across 10+ systems.
- **Containerized ETL pipelines** with Docker and orchestrated them using Kubernetes for scalability and fault tolerance, ensuring robust data processing environments.
- **Complemented** containerization efforts by automating infrastructure provisioning with **Terraform, streamlining** resource management across multi-cloud environments to support scalable ETL operations.
- **Automated** data workflows using Bash scripting in **Linux/Unix** environments, supporting **ETL pipelines** and ensuring efficient multi-cloud integration
- **Led multi-cloud data migration initiatives:** transferred transactional databases to **Azure Data Lake** for analytics and archival data to GCP Big Query for AI/ML workloads, optimizing cost-efficiency by 20%.
- **Implemented** versatile data ingestion workflows, using **Apache Sqoop** for bulk relational database imports and **Apache Flume** for streaming log data into Hadoop, ensuring seamless and reliable data intake.
- **Configured and managed Kafka clusters for real-time data ingestion** and streaming, integrating with Spark Streaming and enabling downstream analytics pipelines.
- **Performed** data transformations and analytics using **PySpark, Spark Core, Spark SQL, Java, and GCP Big Query**, enabling scalable and efficient data processing across systems.
- **Developed** custom MapReduce programs, **Hive UDFs, and HBase** tables to process queries and store large datasets, optimizing workflows for business analytics.
- **Architected** and implemented large-scale BI solutions using **Azure Synapse Analytics, Databricks, GCP Big Query, and Cloud Pub/Sub**, delivering scalable and reliable analytics pipelines.
- **Designed** and implemented **Star and Snowflake schemas**, creating Fact and Dimension tables with tools like Erwin, supporting advanced analytics and reporting.
- **Built interactive dashboards** and reports using **Power BI and Tableau**, enabling actionable insights and strategic decision-making.
- **Automated workflows** using **Apache Airflow DAGs**, orchestrating end-to-end data pipelines for efficient execution and monitoring.
- **Designed MongoDB** schemas with efficient data structures for document modeling, indexing, and tuning, improving database performance and query efficiency.
- **Conducted comprehensive** data analysis, cleaning, and transformation using **Python libraries** (pandas, NumPy) to optimize machine learning models and predictive analytics workflows.
- **Leveraged AWS Glue and S3 for ETL workflows** and scalable data storage, complementing multi-cloud data processing strategies.
- **Utilized** Git for version control and implemented **CI/CD pipelines** using **Jenkins and GitLab**, ensuring streamlined deployments and efficient collaboration across teams.
- **Used Jira** for issue tracking and task management to align with project goals

Environment: Apache Spark, Scala, PySpark, Python (Pandas, NumPy), Hadoop (HDFS, MapReduce, Hive, HBase), Azure (Data Factory, Data Lake, Synapse Analytics, Databricks), GCP (Big Query, Dataflow, Pub/Sub, Cloud Storage), AWS (Glue, S3, EMR, EC2, Redshift, Lambda), Docker, Kubernetes, Terraform, Jenkins, GitLab, Sqoop, Flume, Linux, Bash scripting, Tableau, Power BI, Kafka, Airflow, MongoDB, SQL, PL/SQL, Git, Jira, Agile.

Centene Corporation
Big Data Developer

St. Louis, MO
Sep 2017-Apr 2019

Responsibilities:

- Created and maintained **advanced SQL scripts, indexes, views, and complex queries** to facilitate comprehensive data analysis and extraction.
- Conducted **data mapping, cleansing, and program development to verify and load data**, ensuring the accuracy of converted information from legacy systems.
- Developed diverse data mappings to meet business requirements and project goals effectively.
- Processed source data into staging environments by applying techniques like **data merging, cleansing, and aggregation for effective preprocessing**.
- **Designed and published interactive Power BI dashboards and reports**, enabling streamlined access and visualization through the Power BI service.
- **Established seamless live data connections between Power BI and multiple databases**, enabling automatic updates and query refreshes.
- Used **Power Query within Power BI to pivot and unpivot data models**, ensuring efficient data cleansing and manipulation.
- Imported data into **Power BI from sources such as SQL Server, Oracle, and SQL Azure** for comprehensive reporting.
- **Optimized SQL queries for enhanced performance** and supported data loading operations to improve system efficiency.
- Worked with multiple data formats, **including Parquet, AVRO, Sequence Files, Map Files, and XML**, to handle diverse datasets.
- **Scheduled daily data transfers to S3 buckets and utilized AWS Glue to load data into Redshift while automating processes with AWS Lambda**.
- Managed the continuous integration and deployment of data infrastructure components, including **Apache Spark clusters, Kafka brokers, and Hadoop clusters, to deliver scalable and resilient processing environments**.
- Monitored and improved database performance using tools like **Apache Ambari and Prometheus**, addressing bottlenecks and inefficiencies to ensure system reliability in Big Data environments.
- **Executed test cases across different testing phases**, such as integration and regression testing, to validate system integrity

Environment: SQL, Power BI, Power Query, SQL Server, Oracle, SQL Azure, Parquet, AVRO, Sequence Files, Map Files, XML, Talend, AWS S3, AWS Glue, AWS Lambda, Apache Spark, Kafka, Hadoop, Apache Ambari, Prometheus, Agile Methodology.

EDUCATION

- **Master's in data Analytics and Insights** - Wright State University, Fairborn, Ohio